



Federated Open OnDemand

Robin Karlsson, CSC IT Center for Science, Finland

18.03.2025



Structure

- Briefly about CSC
- Motivation and Context
- EuroHPC Federation Platform
- OOD Features and Requirements
- Challenges in Federated OOD
- Potential Solutions
- Long-term Challenges

Who Am I?

- Robin Karlsson
- Working at CSC with OOD for ~4 years
- Active on the ondemand GitHub repo
 - Largest contribution: Remote file system support in file browser (in 3.0)

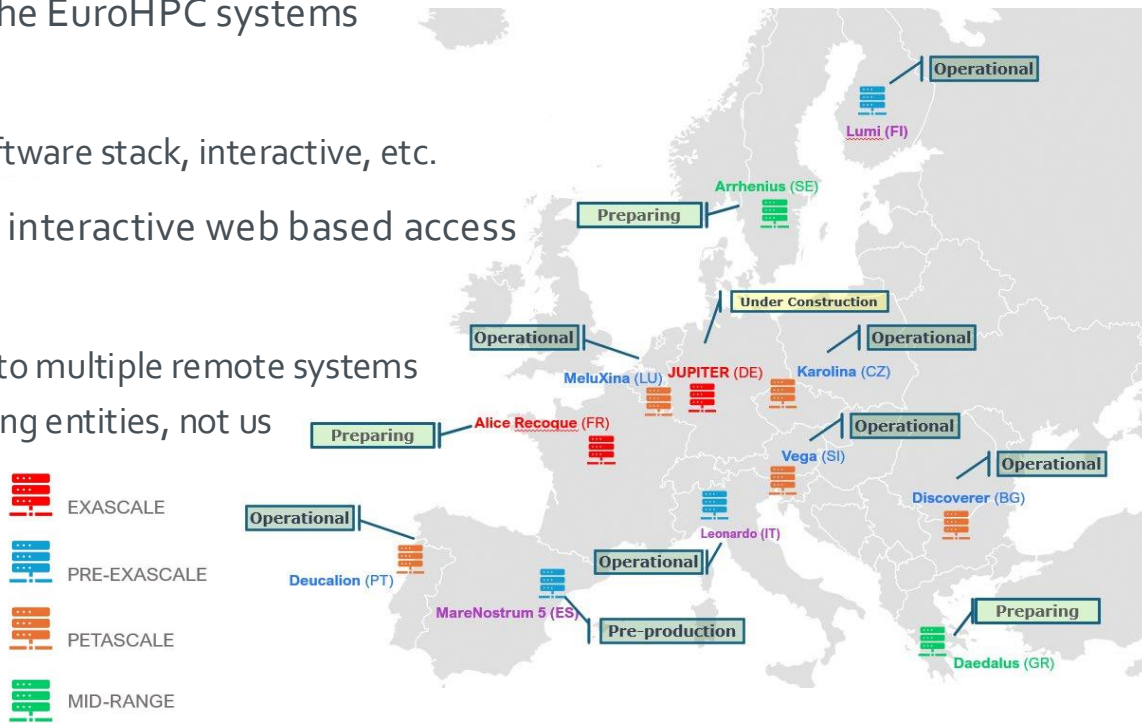
CSC – IT Center for Science

- National, state-owned enterprise
- Provides IT services for academia, research institutes and companies
- Two national supercomputers, Puhti and Mahti
- Hosts the European LUMI supercomputer
 - #8 on the TOP500 list 11/2024
- OOD in use on all 3 supercomputers

Motivation – EuroHPC Federation Platform

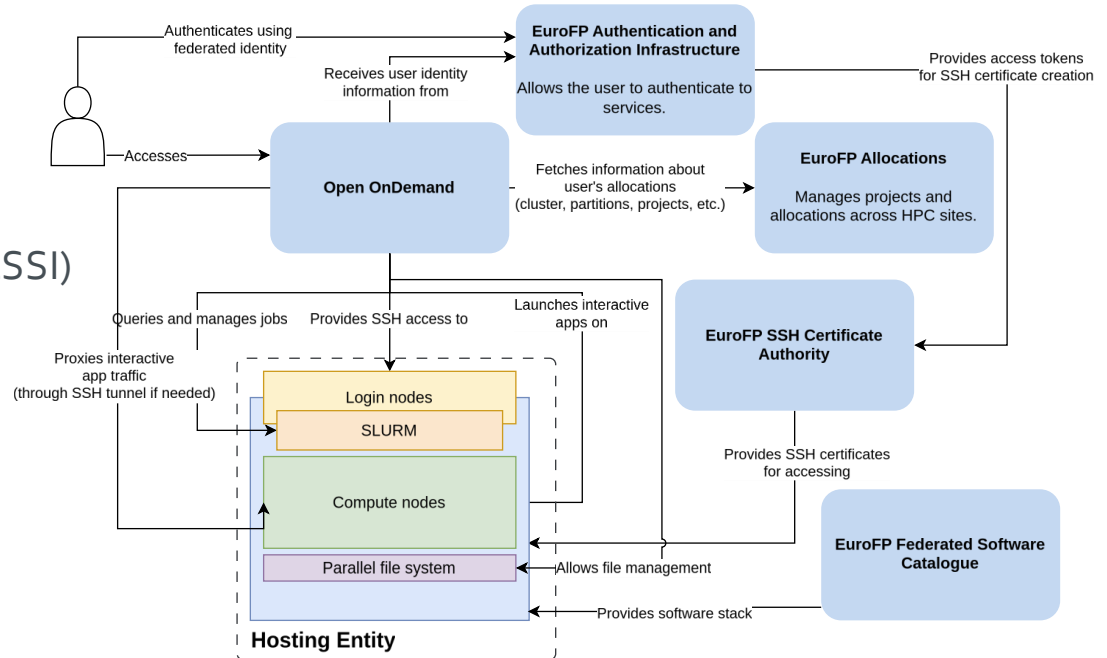
- A platform to federate access to the EuroHPC systems
- Many components
 - Resource allocation, workflows, software stack, interactive, etc.
- Open OnDemand used to provide interactive web based access
- Federated Open OnDemand
 - Single instance of OOD connected to multiple remote systems
 - Remote systems managed by hosting entities, not us

The EuroHPC Ecosystem
2019 - 2024

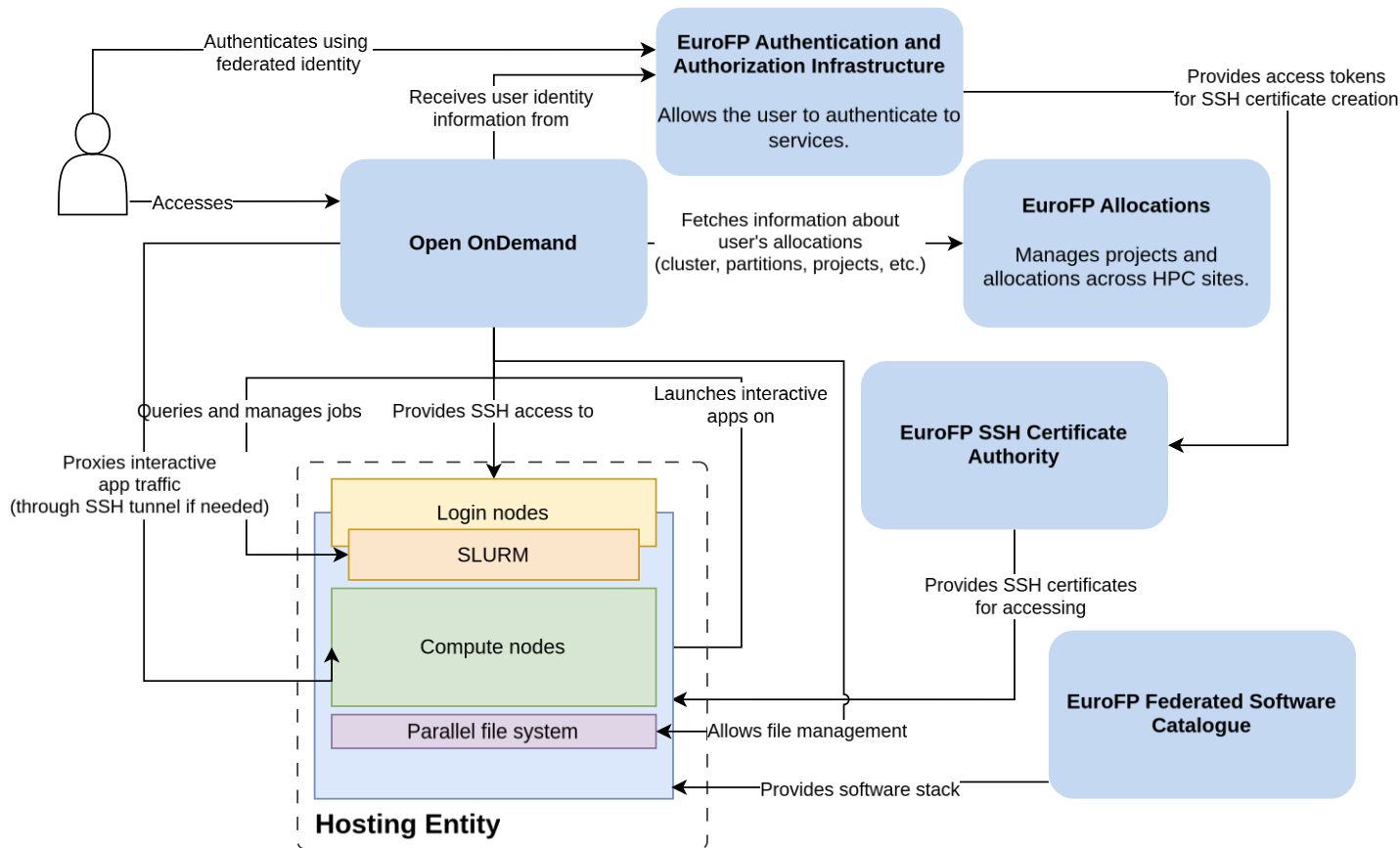


EuroHPC Federation Platform – Foundation

- Several platform components are needed to build Federated OOD
- Federated Identity
- Federated Resource Allocation
- Federated SSH CA
- Federated Software Catalogue (EESSI)



EuroHPC Federation Platform – Foundation

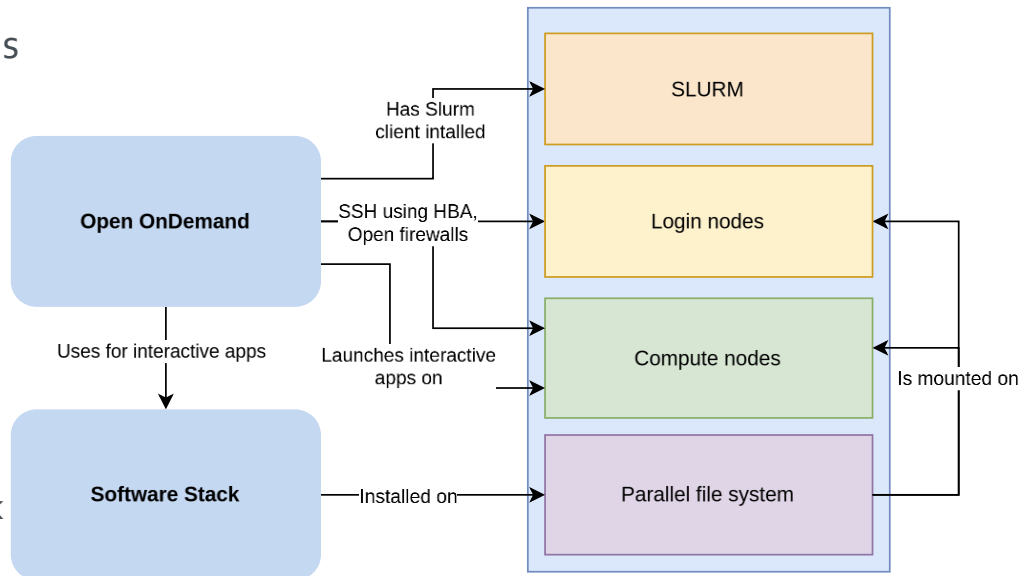


OOD Features and Requirements

- Interactive apps:
 - Scheduler access
 - Cluster file system access
 - Software stack
 - Network access to compute nodes
- File management
 - Cluster file system access
- Shell apps
 - SSH access to login and/or compute nodes
- Job Composer / Project Manager
 - Scheduler access
 - Cluster file system access

OOD Features and Requirements – Normal OOD Setup

- SSH access to login and/or compute nodes
 - HBA configured
- Scheduler access
 - Direct scheduler access on utility node
- Cluster file system access
 - Cluster file system mounted
- Network access to compute nodes
 - Firewall openings in internal cluster network
- Software stack
 - A single software stack on the cluster file system



Federated Open OnDemand

- SSH access
 - Exchange access tokens for SSH certificates (from SSH CA)
- Scheduler access
 - OOD already supports using scheduler through login nodes (SSH)
- Software stack
 - Use EESSI

Federated Open OnDemand – Cluster File System Access

- Mounting cluster file system in the typical way is not possible
- Two other approaches: sshfs and a more custom approach

sshfs

- Mount cluster file system using sshfs on login, unmount on logout

+ Simple

- Lack of flexibility, only SSH
- Potential reliability issues
- Some "bugs", e.g. UID/GID handling

Custom abstraction

- Idea: file browser supports local files ("PosixFile") and files over rclone ("RemoteFile"). Implement "ClusterFile"

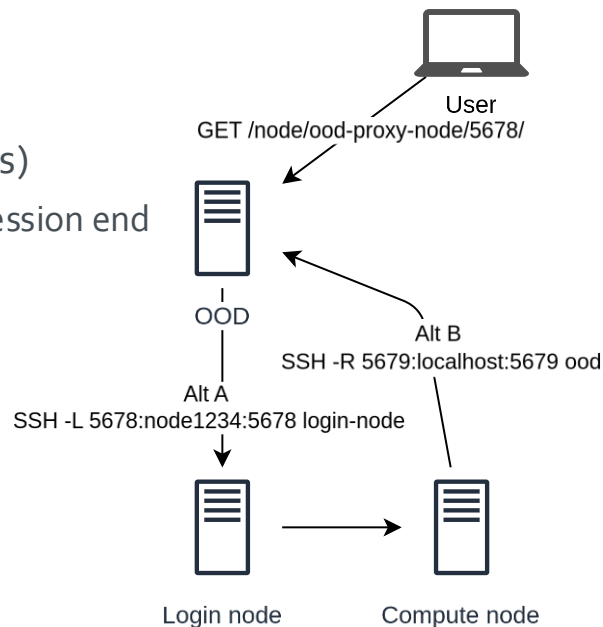
+ Flexible with potential for future improvements

+ Implement support for HPC APIs

- Complex
- Significantly more work

Federated Open OnDemand – Compute Node Network Access

- Firewall openings unlikely
- Two alternatives, both relying on SSH tunnels
- Alt. A: SSH tunnel to the compute nodes (through login nodes)
 - Initiate an SSH tunnel on interactive job app launch, clean up on session end
 - Seemingly easier to implement
- Alt. B: Reverse SSH tunnel from the compute nodes
 - Interactive app job initiates reverse SSH tunnel
 - More difficult to secure



Federated Open OnDemand – Other Concerns

- Performance
 - Can a single instance handle the number of users of 9 supercomputers?
 - Load balancing may be necessary
- Further optimization
 - Extra roundtrip for network traffic in some cases (compared to normal OOD)
 - E.g. user and supercomputer in Spain, OOD in Finland
 - Traffic goes Spain (user) -> Finland -> Spain (supercomputer) and back



Robin Karlsson



facebook.com/CSCfi



twitter.com/CSCfi



linkedin.com/company/csc--it-center-for-science



github.com/CSCfi